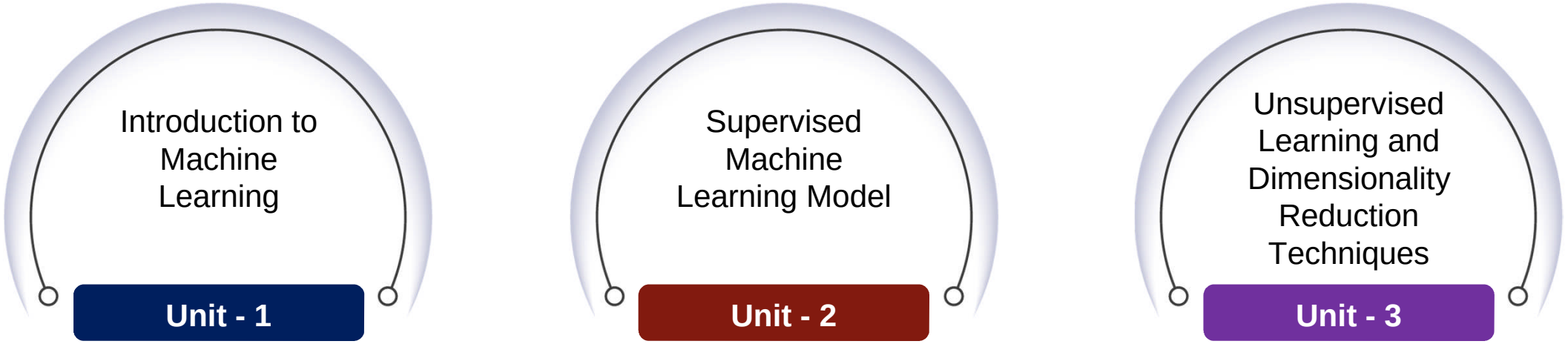
Module - 1

Statistical Modelling & Predictive Analytics – Machine Learning



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Units for Discussion



Unit - 1

Introduction to Machine Learning

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DISCLAIMER

The content is curated from online/offline resources and used for educational purpose only.

Learning Objectives

- About Machine Learning
- Applications
- Types of Platforms
- Categories of Machine Learning
- GUI & Non-GUI Approach
- "Hello world to ML"

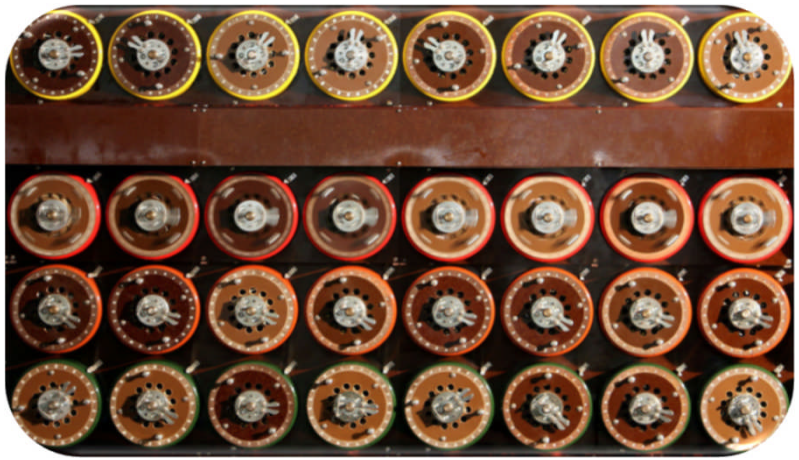


Source :

www.freepik.com/

About Machine Learning

- Concept emerged during WW-II
- Primarily knows as Turing Machine
- Intent to learn encrypted message.
- Accepted as field of science in 1950’s.
- Fundamental in the development of early AI concepts.





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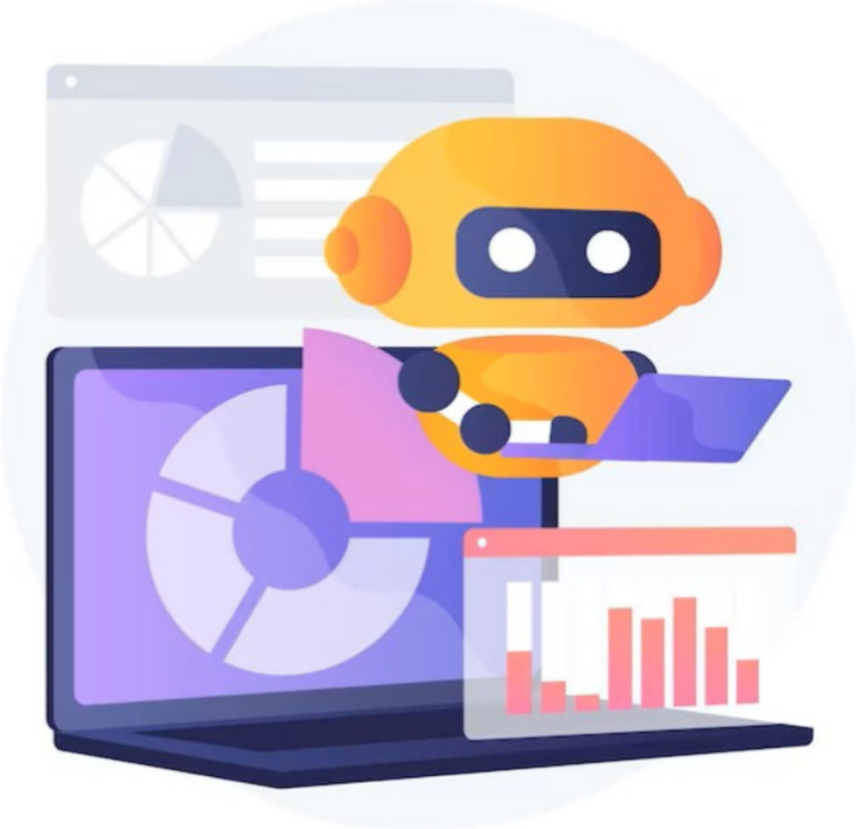
Reference link

Machine Learning: Definition

Early definition of Machine Learning

“Field of study that gives computers the ability to learn without being explicitly programmed”. - Arthur Samuel (1959)

- What do you mean by Explicitly Programmed ?
- So, machine learning algorithms, inspired by the human learning process, iteratively learn from data, and allow computers to find hidden insights.
- These models help us in a variety of tasks, such as object recognition, summarization, recommendation, and so on.



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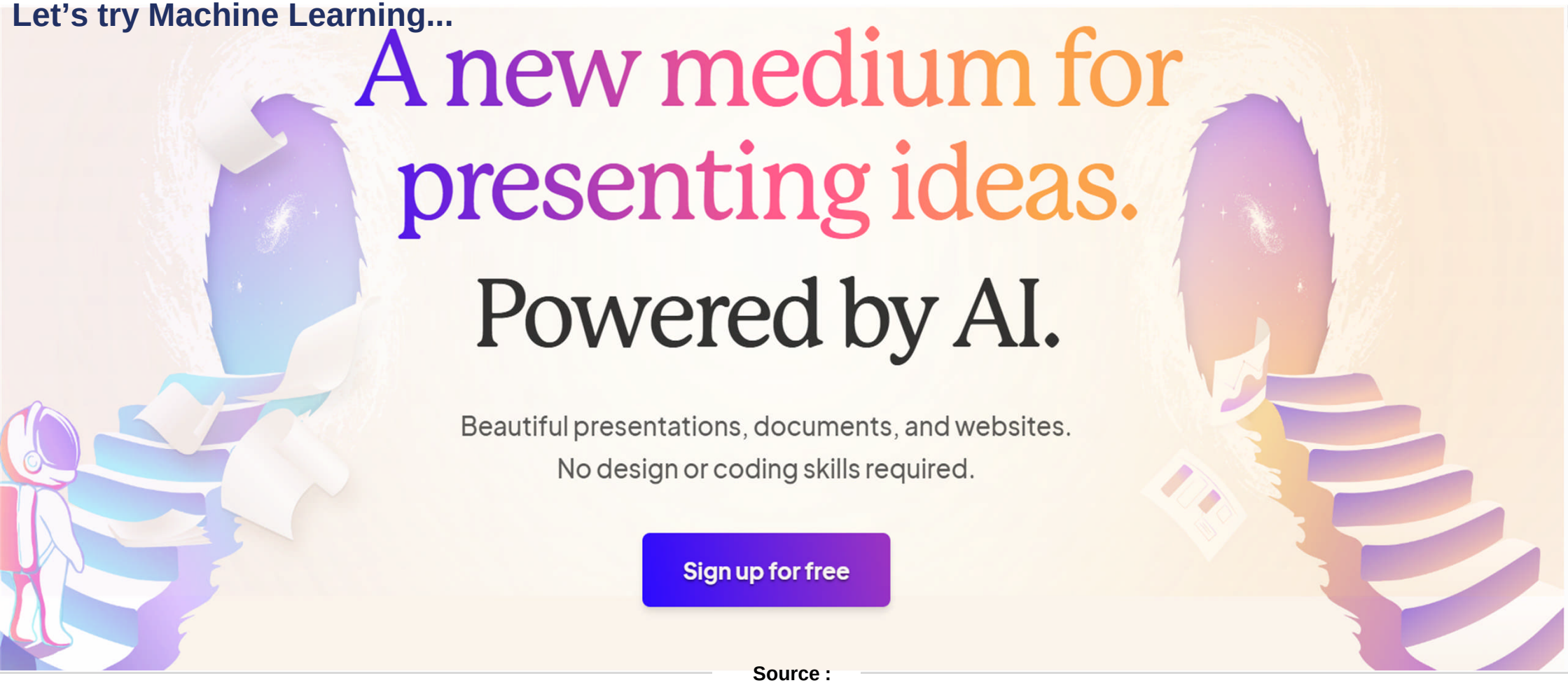
www.freepik.com/

Traditional Programming Vs Machine Learning





Let’s try Machine Learning...



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Reference link

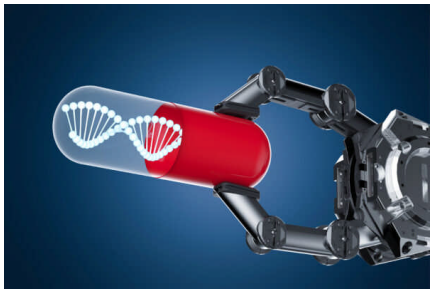
Machine Learning Applications



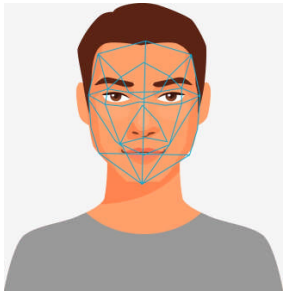
blackbox.ai



Recommendation System



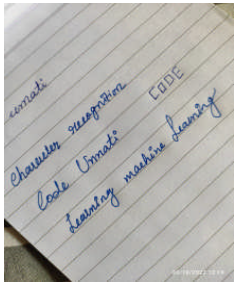
Drug Discovery



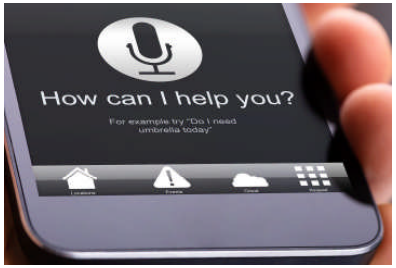
Facial Recognition



Assisted Driving

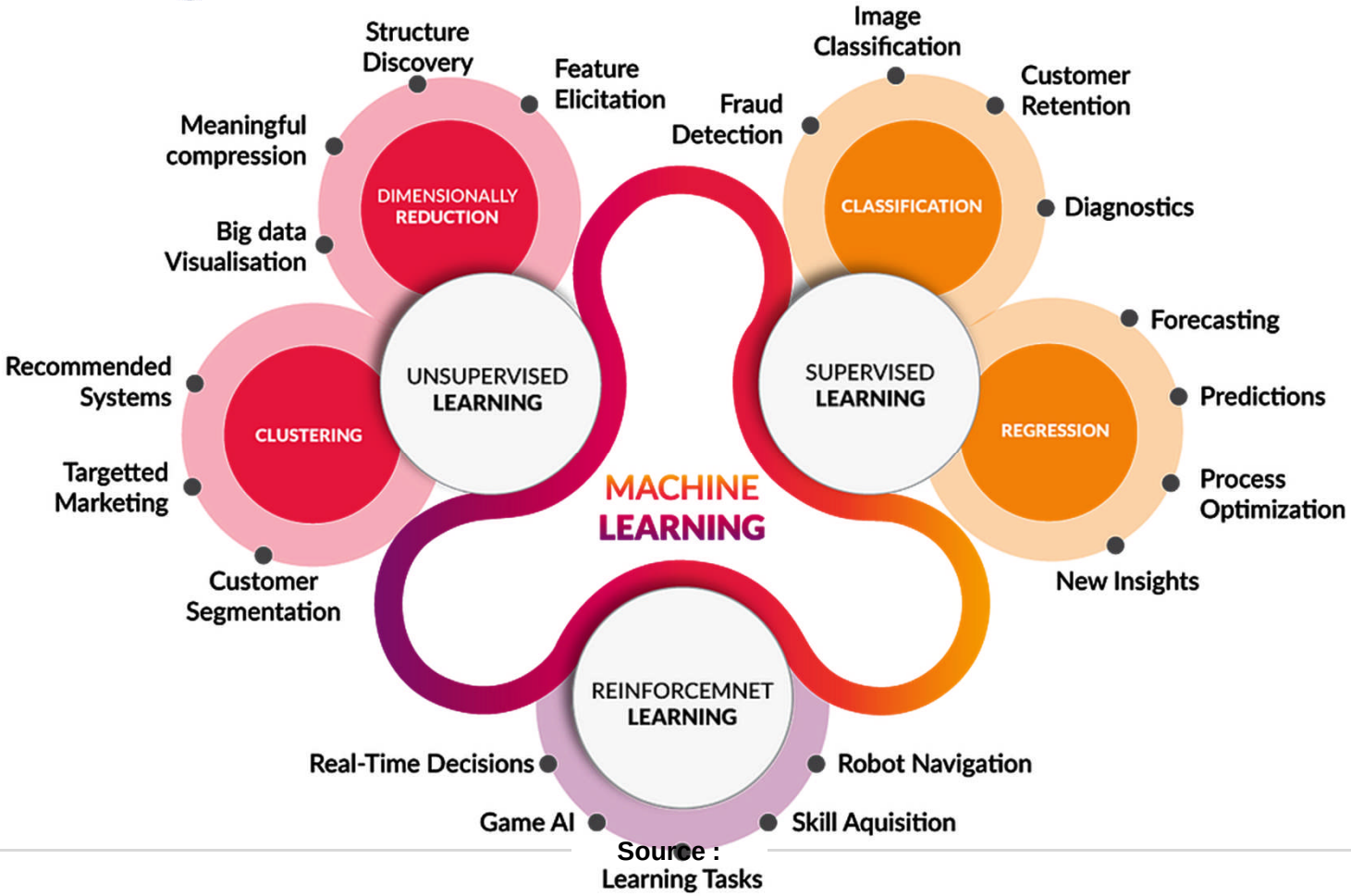


Character Recognition



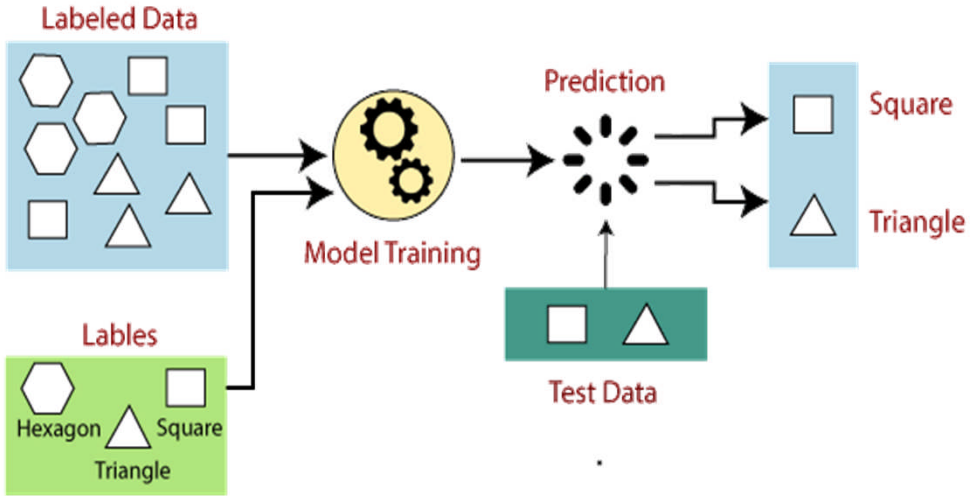
voice-assistant

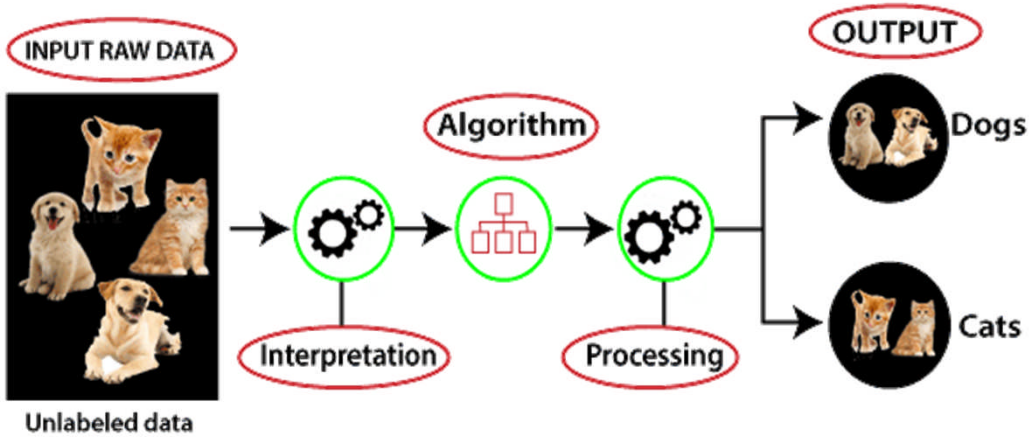
Types of Machine Learning



Reference link

ML Categories: Supervised & Unsupervised





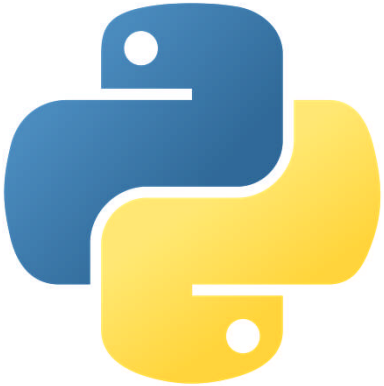
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Supervised Learning

Unsupervised Learning

Coding Platform: Python

- Python is a General-Purpose Programming language that is often applied in scripting roles.
- So, Python is programming language as well as scripting language.
- Python is an Interpreted language
- Simple and easy-to-learn syntax.



python™

Source :

Python

Python - Uses



Source :

Reference link

Anaconda : Introduction

- An easy-to-install collection of high-performance Python libraries
- For managing packages and environments.
- Can use Conda to install over 1.5k packages (including the R language) from the Anaconda public repository.
- More than 20k packages from community channels, such as Conda-forge and bioconda



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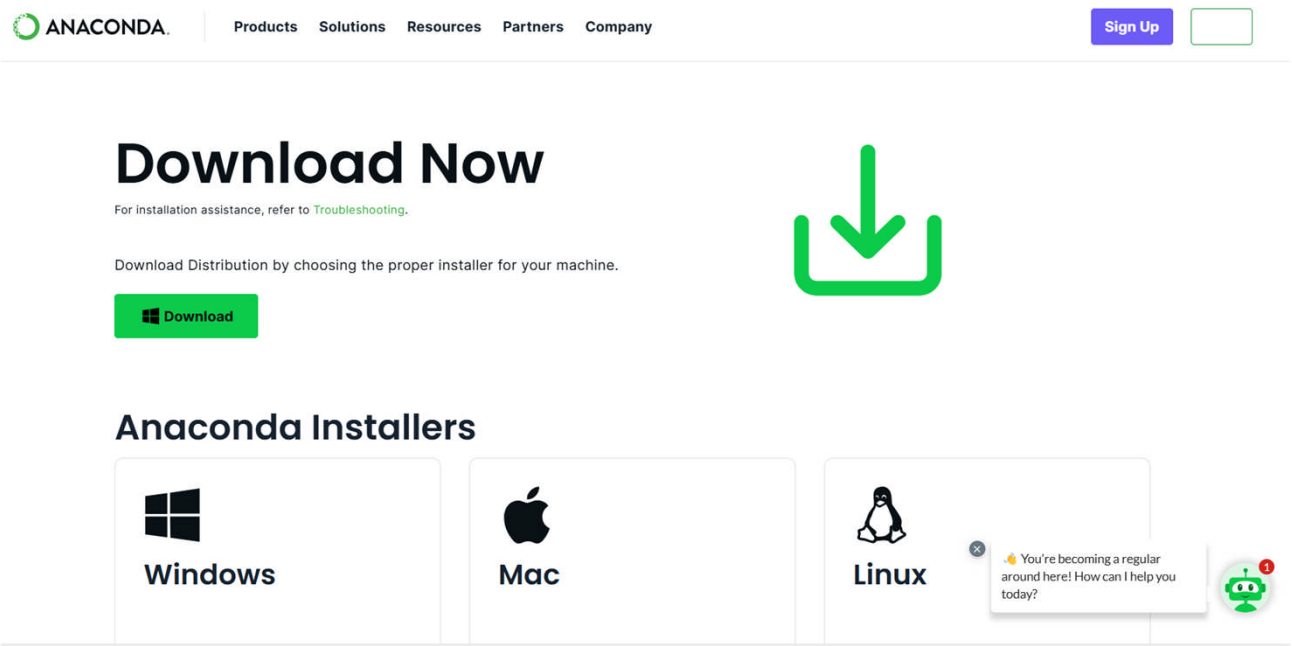
Reference link

Lab - 1

Demonstration of Anaconda Installation

Anaconda Installation

- Visit the following link: [Download Now | Anaconda]
- Scroll down the page and select windows.



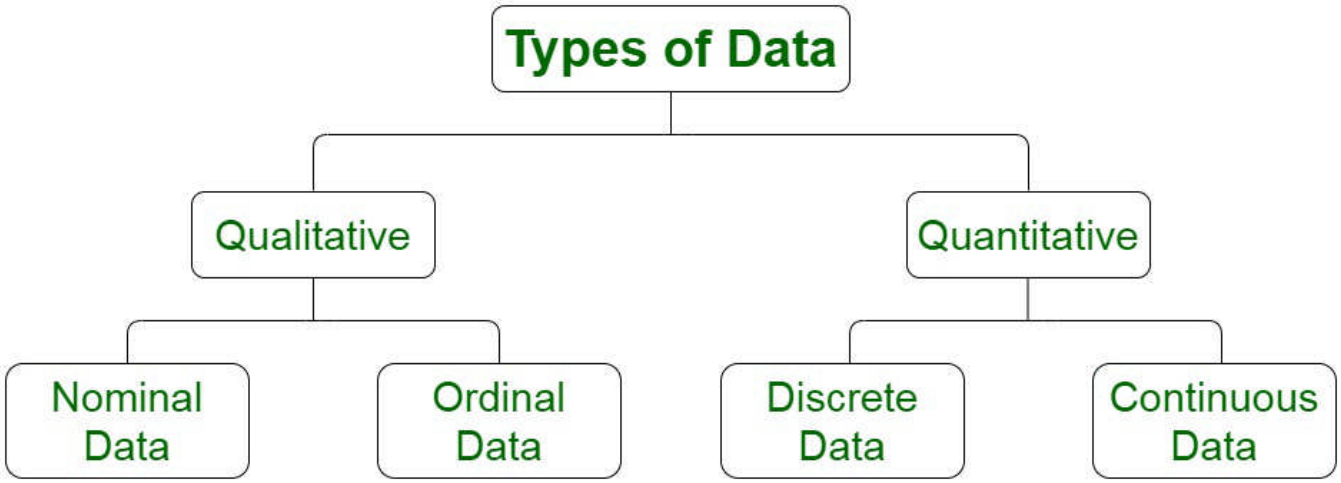
Source :

Reference link

Types of Data

Example

- Nominal – Good, Bad,…
- Ordinal – First, Second….
- Discrete – Student count
- Continuous -- Temperature



Source :

Reference

Requisite libraries: NumPy, Pandas, Matplotlib & Seaborn

NumPy	Pandas	Matplotlib	Seaborn
• Multidimensional arrays and matrices • High-level mathematical functions • **pip install numpy** • **import numpy** • **numpy.__version__**	• Easy data structure • quicker data analysis • Structed & Un-structured data • **pip install pandas** • **import pandas as pd**	• Easy data visualization • Wide variety of plots • Customization and control • **pip install matplotlib** • **Import matplotlib.pyplot as plt**	• Seaborn • Visualization library • Statistical graphics plots • Relational patterns • Used for EDA • **pip install seaborn** • **import seaborn as sns**

Scikit-Learn Library

- A free machine learning library.
- Includes most of the classification, regression and clustering algorithms.
- Works with numerical and scientific libraries, NumPy and SciPy.
- Machine learning pipelined tasks are already in scikit learn.
- Includes pre-processing, feature selection, data splitting, customize algorithms, fitting models, tuning parameters, prediction, evaluation, and exporting the model.
- Has an extended support for Deep learning and cloud services.



Source :

Reference

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Machine Learning Workflow



Source :

www.freepik.com/

Lab - 2

Write a code to differentiate between Apples & Oranges ?

Training Data

Features		Target
Input of classifier		Output of classifier
Weight	Texture	Label
150g	Bumpy	Orange
170g	Bumpy	Orange
140g	Smooth	Apple
130g	Smooth	Apple

```
import sklearn
features = [[140,"smooth"],[130,"smooth"],[150,"bumpy"],[170,"bumpy"]]
labels = ["apples", "apples", "orange", "orange"]
```

Change strings to integers

First 3 Lines of Code !

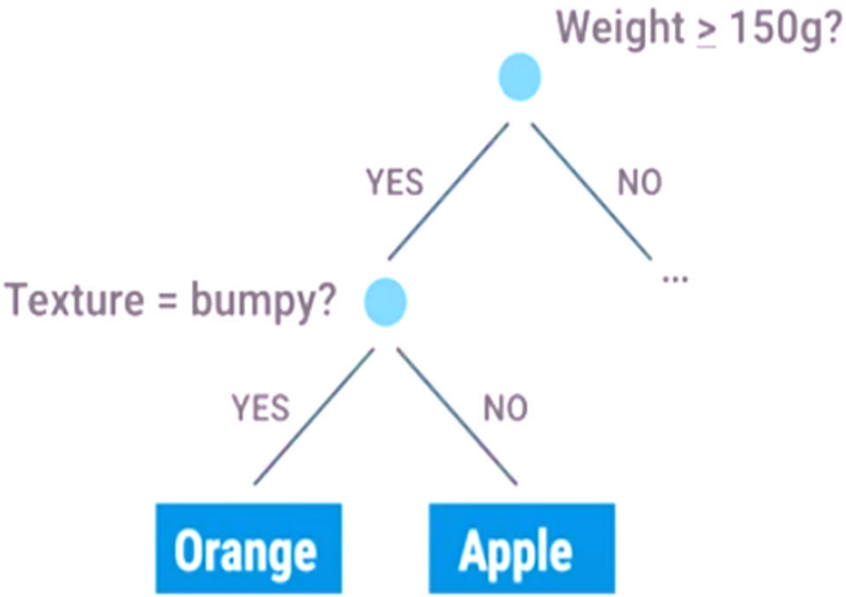
```
import sklearn
features = [[140, 1], [130, 1], [150, 0], [170, 0]]
labels = [0, 0, 1, 1]
```

0: bumpy ; 1: smooth
0: apple ; 1: orange

Classifier

- Box of RULES
- Learning Algorithms are the procedure that creates RULES, by finding patterns in your training DATA.
- Ex: It creates RULE that heavier fruit is more likely to be an orange!

Decision Tree



Machine Learning……Final Code

```
from sklearn import tree
features = [[140, 1], [130, 1], [150, 0], [170, 0]]
labels = [0, 0, 1, 1]
clf = tree.DecisionTreeClassifier()
clf = clf.fit(features, labels)
print(clf.predict([[150, 0]]))
```

0: bumpy ; 1: smooth

0: apple ; 1: orange

Classifier gets trained on input data

Need for Model Evaluation

- Built on a subset of the total data, termed as training data, and they are used to predict on new data that is not part of this training subset.
- If a model is totally adapted to its training data, it would fail to predict accurately any new data (**Overfitting**).
- If model is too general, it would predict poorly on cases (**Underfitting**).
- A good model should be perfectly balanced to avoid **both**.
- By holding out part of the data from the training set and evaluating model with this subset of test data.

ML: Problem Types

Problem types	Algorithms
Regression	Linear regression, K-NN, random forest, neural networks
Classification	Logistic regression, random forest, K-NN, gradient boosting classifier, neural networks
Clustering	K-Means, DBSCAN, Hierarchical clustering, Gaussian mixture models, BIRCH
Time-series forecasting	ARIMA, SARIMA, LSTM, Exponential smoothing, Prophet, GARCH, TBATS, Dynamic linear models
Anomaly detection	Isolation Forest, Minimum covariance determinant, Local outlier factor, One-class SVM
Recommendation	Content-based and collaborative filtering machine learning methods
Data generation	Generative adversarial network (GAN), Hidden Markov models

Conclusion

- In the first section, we see definition of Machine Learning, i.e. ML algorithms learn patterns and relationships within the data and use this knowledge to make predictions, decisions, or automate tasks.
- After that we have seen Machine learning techniques that can be broadly categorized into three main types i.e. Supervised Learning , Unsupervised Learning and Reinforcement Learning
- We have seen differences between using a Graphical User Interface (GUI) and bare coding (writing code directly) for Machine Learning (ML) hands-on work



Source :

www.freepik.com/

References

- https://en.wikipedia.org/wiki/Git
- https://en.wikipedia.org/wiki/GitHub
- https://medium.com/machine-learning-101
- https://medium.com/@randylaosat/a-beginners-guide-to-machine-learning-dfadc19f6caf
- https://sml.csa.iisc.ac.in/Courses/Spring21/E0_270/pdfs/1.pdf
- https://www.geeksforgeeks.org/difference-between-git-and-github/
- https://link.springer.com/book/10.1007/978-3-030-81935-4
- https://ai.stanford.edu/~nilsson/MLBOOK.pdf
- https://mitpress.mit.edu/9780262043793/introduction-to-machine-learning/

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Let’s Start

Quiz

1. **Which type of Machine Learning involves learning from a labeled dataset to make**

a) Unsupervised Learning

b) Supervised Learning

c) Semi-Supervised Learning

d) Reinforcement Learning



Answer: B

Supervised Learning

Quiz

2. Which type of Machine Learning Library use for Prediction of based on past data

a) Pandas

b) Numpy

c) Sci-kit Learn (skleran)

d) CV



Answer: C

Sci-kit Learn (skleran)

Quiz

3. Which type of learning is primarily used to find patterns or groupings in data without any labels?

a) Supervised Learning

b) Unsupervised Learning

c) Semi-supervised Learning

d) Reinforcement Learning



Answer: B

Unsupervised Learning

Quiz

4. What is the main goal of Unsupervised Learning?

a) To make predictions based on labelled data

b) To learn from rewards and punishments

c) To find hidden patterns and structures in unlabelled data

d) To improve the accuracy of existing models



Answer: C

To find hidden patterns and structures in unlabelled data

Quiz

5. Which Python Library Use for Data Visualization

a) Line plot

b) Seaborn

c) Matplotlib

d) Scatterplot



Answer: B and C

Seaborn and Matplotlib

Thank You